By repeating the same calculation as (22) for $\boldsymbol{y}_2$, we get:

$$\boldsymbol{w}_j^\intercal \boldsymbol{y}_2 + b_j = (1 - 2\gamma_j) y_j(\boldsymbol{c}). \tag{28}$$

By (21) it follows that (again with $j \neq i$)

$$\begin{aligned} & \boldsymbol{w}_j^\intercal (\boldsymbol{y}_1 - \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}) = \boldsymbol{w}_j^\intercal (\boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{y}_2) \\ \Rightarrow \quad & \boldsymbol{w}_j^\intercal \boldsymbol{y}_1 + b_j - \boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)} - b_j = \boldsymbol{w}_j^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_j - \boldsymbol{w}_j^\intercal \boldsymbol{y}_2 - b_j \\ \Rightarrow \quad & (1 - 2\lambda_j) y_j(\boldsymbol{c}) - y_j(\boldsymbol{c}) = y_j(\boldsymbol{c}) - (1 - 2\gamma_j) y_j(\boldsymbol{c}) \\ \Rightarrow \quad & 1 - 2\lambda_j - 1 = 1 - 1 + 2\gamma_j \\ \Rightarrow \quad & \lambda_j + \gamma_j = 0. \end{aligned} \tag{29}$$

Clearly, since λ_j and γ_j are convex combinations and thus non-negative, (29) implies that $\lambda_j = \gamma_j = 0$.

By repeating this process for all $j \neq i$, we get that $\lambda_j = \gamma_j = 0$ for all $j \neq i$, and therefore $\lambda_i = \gamma_i = 1$. From this, it follows that $\boldsymbol{y}_1 = \boldsymbol{z}_{\boldsymbol{c}}$ and $\boldsymbol{y}_2 = \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}$.

This means that

$$\boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)} + y_i(\boldsymbol{c}) \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i = \boldsymbol{z}_{\boldsymbol{c}} \quad \text{and} \quad \boldsymbol{z}_{\boldsymbol{c}} - y_i(\boldsymbol{c}) \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i = \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}, \tag{30}$$

and therefore the differences between $\boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}$ are independent of other concept variations. Because of that, we can write any datapoint $\boldsymbol{z}_{\boldsymbol{c}}$ as a sum of concept-specific values $\boldsymbol{u}_{i,c_i} (c_i \in [2])$. For instance, if we fix $\mathbf{0} = (0, \ldots, 0) \in [2]^k$, we can express $\boldsymbol{z}_{\boldsymbol{c}}$ as, for example (up to a global linear shift per concept)

$$\begin{aligned} \boldsymbol{u}_{i,0} = \boldsymbol{z}_{\mathbf{0}}/k, \quad \boldsymbol{u}_{i,1} = \boldsymbol{z}_{\mathbf{0}}/k + \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i, \\ \boldsymbol{z}_{\boldsymbol{c}} = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i}, \end{aligned} \tag{31}$$

from here, we can write any datapoint $\boldsymbol{z}_{\boldsymbol{c}}$ as

$$\boldsymbol{z}_{\boldsymbol{c}} = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i} = \sum_{i \in [k]: c_i = 0}^{k} \boldsymbol{z}_{\mathbf{0}}/k + \sum_{i \in [k]: c_i = 1}^{k} (\boldsymbol{z}_{\mathbf{0}}/k + \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i) = \boldsymbol{z}_{\mathbf{0}} + \sum_{i \in [k]: c_i = 1}^{k} \frac{2}{||\boldsymbol{w}_i||^2} \boldsymbol{w}_i, \tag{32}$$

which establishes linearity.

Orthogonality. First, note that by invariance (Lemma 3) it holds that for any concept i, changes in concept values other than i do not affect the prediction of concept i:

$$\boldsymbol{w}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_i = \boldsymbol{w}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)} + b_i \tag{33}$$

But by linear factorization (31) it follows that

$$\begin{aligned} & \boldsymbol{w}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}} + b_i = \boldsymbol{w}_i^\intercal \boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)} + b_i \\ \Rightarrow \quad & \boldsymbol{w}_i^\intercal (\boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{z}_{\boldsymbol{c}(j \to \bar{c}_j)}) = 0 \\ \Rightarrow \quad & \boldsymbol{w}_i^\intercal \left(\boldsymbol{u}_{j,c_j} - \boldsymbol{u}_{j,\bar{c}_j} \right) = 0 \\ \Rightarrow \quad & \boldsymbol{w}_i^\intercal \left(\frac{2}{||\boldsymbol{w}_j||^2} \boldsymbol{w}_j \right) = 0 \\ \Rightarrow \quad & \boldsymbol{w}_i^\intercal \boldsymbol{w}_j = 0. \end{aligned} \tag{34}$$

Then,

$$(\boldsymbol{u}_{i,c_i} - \boldsymbol{u}_{i,\bar{c}_i})^\intercal (\boldsymbol{u}_{j,c_j} - \boldsymbol{u}_{j,\bar{c}_j}) \propto \boldsymbol{w}_i^\intercal \boldsymbol{w}_j = 0. \tag{35}$$

More generally, orthogonality of one concept holds against the span of other concepts as well. For $\{\alpha_j \in \mathbb{R}\}_{j \neq i}$ it follows that

$$(\boldsymbol{u}_{i,c_i} - \boldsymbol{u}_{i,\bar{c}_i})^\intercal \left(\sum_{j \neq i} \alpha_j (\boldsymbol{u}_{j,c_j} - \boldsymbol{u}_{j,\bar{c}_j}) \right) \propto \boldsymbol{w}_i^\intercal \left(\sum_{j \neq i} \alpha_j \boldsymbol{w}_j \right) = 0, \tag{36}$$

and therefore orthogonality holds against the span of other concepts differences. □

D. Sufficiency of linear factorization for compositional generalization (proof of Proposition 2)

This section complements the main text's necessity results by showing a converse: under the same linear-head setting, linearly factored representations are sufficient to obtain compositional generalization under our desiderata.

We discuss two cases. First, in the binary setting, we show that linear factorization together with cross-concept orthogonality makes stable transfer automatic for GD+CE, even from small but diverse training supports (Section D.1). Second, in the general multi-valued setting we discuss the general case when the underlying features are linear, but not necessarily orthogonal across concepts, and the learning algorithm is not necessarily GD+CE (Section D.2). There, training with GD+CE does not necessarily satisfy the desiderata, but in principle classifiers can be constructed that do, precisely because the factors can be recovered.

D.1. Binary valued-case: sufficiency of SVMs

In this section, we show that linear factorization together with cross-concept orthogonality makes stable transfer automatic for GD+CE, even from small but diverse training supports. Since concepts are binary-valued, we can represent the readout for each concept by a single affine separator with parameters $(\boldsymbol{w}_i, b_i)$, and we work with these parameters throughout (in the same way as Section C).

Proposition 4 (Linear factorization implies compositional generalization)**.** Consider the binary grid $\mathcal{C}_i = \{0, 1\}$ with representations $\mathcal{Z} = \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k\} \subset \mathbb{R}^d$. Assume there exist $\{\boldsymbol{u}_{i,0}, \boldsymbol{u}_{i,1} \in \mathbb{R}^d\}_{i=1}^k$ such that for every $\boldsymbol{c} \in [2]^k$:

1. *(Linearity)* $\boldsymbol{z_c} = \sum_{i=1}^k \boldsymbol{u}_{i,c_i}$.
2. *(Cross-concept orthogonality)* $(\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}) \perp (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0})$ for all $i \neq j$.

Then for any training set $T \subseteq [2]^k$ satisfying either of the following conditions:

1. Any set with $|T| = 2^{k-1} + 1$, or
2. A cross dataset $\mathcal{D}^{\boldsymbol{c}}$ (i.e. the center $\boldsymbol{c}$ and all one-value flips from $\boldsymbol{c}$, Definition 6) with $|T| = 1 + k$.

It holds that gradient descent + cross-entropy loss trained on T satisfy the desiderata on the entire grid: they recover every concept value on all $\boldsymbol{z_c}$ (transferability) and are invariant across valid T (stability).

Proof. Throughout the proof we use the standard result that the GD+CE converges to the max-margin SVM solution in the binary-class case (Soudry et al., 2024) and interpret the optimal solution as a weight vector $\boldsymbol{w}_i$ and bias b_i that separates the two classes.

Full dataset case.

First, we establish the exact weights produced by the linear probes.

For that, note that for any concept $i \in [k]$, the SVM solution is proportional to the shortest segment between the convex hulls of the points in the two classes, denoted side by side as

$$\mathcal{Y}_- := \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k,\ c_i = 0\}, \qquad \mathcal{Y}_+ := \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k,\ c_i = 1\}. \tag{37}$$

with the proportionality constant $\frac{2}{\|\boldsymbol{w}_i\|^2}$ (Bennett & Bredensteiner, 2000). Equivalently, there exist convex combinations $\boldsymbol{y}_- \in \operatorname{conv}(\mathcal{Y}_-)$ and $\boldsymbol{y}_+ \in \operatorname{conv}(\mathcal{Y}_+)$ such that $\boldsymbol{w}_i$ is parallel to the shortest segment $\boldsymbol{y}_+ - \boldsymbol{y}_-$, where

$$\boldsymbol{y}_- = \sum_{\boldsymbol{c} \in [2]^k, c_i = 0} \gamma_{\boldsymbol{c}} \boldsymbol{z_c}, \qquad \boldsymbol{y}_+ = \sum_{\boldsymbol{c} \in [2]^k, c_i = 1} \lambda_{\boldsymbol{c}} \boldsymbol{z_c}, \qquad \boldsymbol{\lambda}, \boldsymbol{\gamma} \succeq 0, \qquad \sum_{\boldsymbol{c} \in [2]^k, c_i = 1} \lambda_{\boldsymbol{c}} = 1, \qquad \sum_{\boldsymbol{c} \in [2]^k, c_i = 0} \gamma_{\boldsymbol{c}} = 1. \tag{38}$$

Importantly, these convex combinations are the support vectors of the SVM solution and correspond to the shortest distance between the two classes. To proceed, we first write the difference between any two points in the convex hulls and then lower bound the norm of the difference and arrive at the exact weight vector.

Note that for any $\boldsymbol{y}_+$ and $\boldsymbol{y}_-$ their difference can be written as:

$$\begin{aligned}
\boldsymbol{y}_+ - \boldsymbol{y}_- &= \sum_{\boldsymbol{c} \in [2]^k : c_i = 1} \lambda_{\boldsymbol{c}} \boldsymbol{z_c} - \sum_{\boldsymbol{c} \in [2]^k : c_i = 0} \gamma_{\boldsymbol{c}} \boldsymbol{z_c} && (40)\\
&= \sum_{\boldsymbol{c} \in [2]^k : c_i = 1} \lambda_{\boldsymbol{c}} \left(\sum_{j \neq i} \boldsymbol{u}_{j,c_j} + \boldsymbol{u}_{i,1} \right) - \sum_{\boldsymbol{c} \in [2]^k : c_i = 0} \gamma_{\boldsymbol{c}} \left(\sum_{j \neq i} \boldsymbol{u}_{j,c_j} + \boldsymbol{u}_{i,0} \right) && (41)\\
&= \sum_{\boldsymbol{c} \in [2]^k : c_i = 1} \lambda_{\boldsymbol{c}} \boldsymbol{u}_{i,1} - \sum_{\boldsymbol{c} \in [2]^k : c_i = 0} \gamma_{\boldsymbol{c}} \boldsymbol{u}_{i,0} + \sum_{\boldsymbol{c} \in [2]^k : c_i = 1} \lambda_{\boldsymbol{c}} \sum_{j \neq i} \boldsymbol{u}_{j,c_j} - \sum_{\boldsymbol{c} \in [2]^k : c_i = 0} \gamma_{\boldsymbol{c}} \sum_{j \neq i} \boldsymbol{u}_{j,c_j} && (42)\\
&= \boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0} + \sum_{\boldsymbol{c} \in [2]^k : c_i = 1} \lambda_{\boldsymbol{c}} \sum_{j \neq i} \boldsymbol{u}_{j,c_j} - \sum_{\boldsymbol{c} \in [2]^k : c_i = 0} \gamma_{\boldsymbol{c}} \sum_{j \neq i} \boldsymbol{u}_{j,c_j}, && (44)
\end{aligned}$$

where the last equality uses the fact that the sum runs over all concept combinations, and therefore sums to 1. Note that the last term in (44) can be written as

$$\sum_{\boldsymbol{c}\in[2]^k:c_i=1} \lambda_{\boldsymbol{c}} \sum_{j\neq i} \boldsymbol{u}_{j,c_j} - \sum_{\boldsymbol{c}\in[2]^k:c_i=0} \gamma_{\boldsymbol{c}} \sum_{j\neq i} \boldsymbol{u}_{j,c_j} \tag{45}$$

$$= \sum_{\boldsymbol{c}\in[2]^k:c_i=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) \left(\sum_{j\neq i} \boldsymbol{u}_{j,c_j} \right) \tag{46}$$

$$= \sum_{j\neq i} \sum_{\boldsymbol{c}\in[2]^k:c_i=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) \boldsymbol{u}_{j,c_j} \tag{47}$$

$$= \sum_{j\neq i} \left(\left(\sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) \right) \boldsymbol{u}_{j,1} + \left(\sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=0} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) \right) \boldsymbol{u}_{j,0} \right), \tag{48}$$

but since $\sum_{\boldsymbol{c}\in[2]^k:c_i=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) = 0$, it follows that for any $j \neq i$,

$$\sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) + \sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=0} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) = 0 \tag{49}$$

$$\Rightarrow \sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) = - \sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=0} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)}) \tag{50}$$

We thus denote $\Delta_j := \sum_{\boldsymbol{c}\in[2]^k:c_i=1,c_j=1} (\lambda_{\boldsymbol{c}} - \gamma_{\boldsymbol{c}(i\to 0)})$. The full expression of (40) can be written compactly as

$$\boldsymbol{y}_+ - \boldsymbol{y}_- = \boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0} + \sum_{j\neq i} \Delta_j \boldsymbol{u}_{j,1} - \Delta_j \boldsymbol{u}_{j,0} \tag{51}$$

$$= \boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0} + \sum_{j\neq i} \Delta_j (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}). \tag{52}$$

Recall that by assumption, $\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0} \perp \boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}$ for all $j \neq i$, so it follows that

$$(\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0})^\intercal \left(\sum_{j\neq i} \Delta_j (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}) \right) = \sum_{j\neq i} \Delta_j (\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0})^\intercal (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}) = 0. \tag{53}$$

Therefore, $\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0} \perp \sum_{j\neq i} \Delta_j (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0})$. This allows us to apply the Pythagorean theorem when computing the distance between $\boldsymbol{y}_+$ and $\boldsymbol{y}_-$:

$$\|\boldsymbol{y}_+ - \boldsymbol{y}_-\|^2 = \|\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}\|^2 + \left\| \sum_{j\neq i} \Delta_j (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}) \right\|^2 \tag{54}$$

$$\geq \|\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}\|^2. \tag{55}$$

Thus, any two points in their respective convex hulls are at least as far apart as the distance between $\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}$. We make use of this result in computing the SVM solution by picking two points $\boldsymbol{y}_+$ and $\boldsymbol{y}_-$ that have exactly the shortest possible distance between them, thus establishing them as the support vectors. Conveniently, any two "counterfactual" points do: for any $\boldsymbol{c} \in [2]^k$ by picking $\boldsymbol{y}_+ = \boldsymbol{z}_{\boldsymbol{c}(i\to 1)}$ and $\boldsymbol{y}_- = \boldsymbol{z}_{\boldsymbol{c}(i\to 0)}$, we have that

$$\|\boldsymbol{z}_{\boldsymbol{c}(i\to 1)} - \boldsymbol{z}_{\boldsymbol{c}(i\to 0)}\|^2 = \|\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}\|^2.$$

Since this computation is independent of the particular choice of the concept i, it holds for any concept. As such, we can write the weight vector $\boldsymbol{w}_i$ as

$$\boldsymbol{w}_i = \frac{2}{||\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}||^2} (\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}). \tag{56}$$

To show that classification works, note that

$$\boldsymbol{z}_{\boldsymbol{c}} = \sum_{j=1}^{k} \left(\widetilde{\boldsymbol{u}}_{j,c_j} + \bar{\boldsymbol{u}}_j \right) = \sum_{j}^{k} \widetilde{\boldsymbol{u}}_{j,c_j} + \sum_{j}^{k} \bar{\boldsymbol{u}}_j, \tag{57}$$